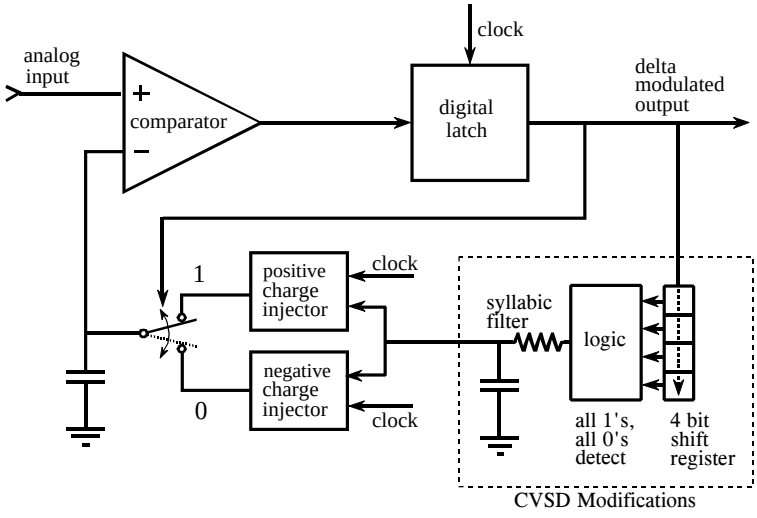

FIGURE 3-18
CVSD modulation block diagram. A logic circuit is added to the basic delta modulator to improve the slew rate.

capacitor is *decreased* when the circuit's output is a digital *one*, and *increased* when it is a digital *zero*. As the input signal increases and decreases in voltage, it tries to raise and lower the voltage on the capacitor. This change in voltage is detected by the comparator, resulting in the charge injectors producing a *counteracting* charge to keep the capacitor at zero volts.

If the input voltage is positive, the digital output will be composed of more ones than zeros. The excess number of ones being needed to generate the *negative* charge that cancels with the *positive* input signal. Likewise, if the input voltage is negative, the digital output will be composed of more zeros than ones, providing a net positive charge injection. If the input signal is equal to zero volts, an equal number of ones and zeros will be generated in the output, providing an overall charge injection of zero.

The relative number of ones and zeros in the output is now related to the *level* of the input voltage, not the *slope* as in the previous circuit. This is much simpler. For instance, you could form a 12 bit ADC by feeding the digital output into a counter, and counting the number of *ones* over 4096 clock cycles. A digital number of 4095 would correspond to the maximum positive input voltage. Likewise, digital number 0 would correspond to the maximum negative input voltage, and 2048 would correspond to an input voltage of zero. This also shows the origin of the name, *delta-sigma*: delta modulation followed by summation (sigma).

The ones and zeros produced by this type of delta modulator are very easy to transform back into an analog signal. All that is required is an analog low-pass filter, which might be as simple as a single RC network. The high